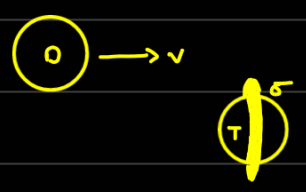
Basic Nuclear Collisions

Cross-section:

If D is an incident particle on T, the area where D can impact to cause a nuclear reaction is the cross-section. This interaction is called a collision.

It is not wrong to imagine the nucleus being surrounded by a force field and the cross-section is the projection of that spherical force field ⊥ to the velocity vector.



The cross section for fusion is on the order of a nuclear diameter. Due to the nature of the strong force, particles passing by will experience a very weak force making the diagram accurate to an extent.

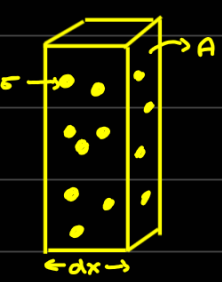
This cross-section is known as a "hard-sphere" cross-section. Actual fusion cross-sections are more complicated but this should suffice for now. Assume σ is const. and known.

Q: "How much" of the particle should cross through the cross-section in order to initiate the reaction?

Mean free path:

The probability of a fusion reaction happening depends greatly on σ. Small σ will result in low probability, large σ will result in high probability.

We can tie cross sections to probability using the concept of mean free path.



Look at this volume (V) of nuclei, if n_1 is the number density of target particles we have.

$$M_1 = n_1 V = n_1 A dx$$

Remember! We are dealing with nuclei here. These are very small in nature and are interspersed through the volume. It is thus a good approximation to say that the σ's do not overlap from the incident particle's POV.

$$dF = \frac{\sigma M_1}{A} = \frac{\sigma n_1 A dx}{A} = \sigma n_1 dx$$ → Fraction of volume blocked by the target nuclei.

We can also write: $\frac{dF}{dx} = \sigma n_1$, This can be interpreted as the probability of nuclear collision per unit length.

Q: [illegible] effects into pair conservation?

Q: Why are we not bringing in Quantum Mechanical effects into this calculation?

If we now assume that a similar cross-section of incident particles are coming in with a velocity v

$$\Gamma = \frac{N_2}{A\,dt} = \frac{n_2 A v\,dt}{A\,dt} = n_2 v$$ → Γ represents the flux of incoming particles. i.e Total no. of particles passing an area A p.u A p.u t

The portion of this flux having a nuclear collision:

$$d\Gamma = dF.\Gamma = \sigma n_1 \Gamma\, dx$$

You can also think about $-d\Gamma$ as the decrease in the total flux of incident particles as you move through a distance dx.

$$-d\Gamma = \Gamma(x+dx) - \Gamma(x) = d\Gamma$$

$$\therefore d\Gamma = -\sigma n_1 \Gamma\, dx$$

This can be solved under the assumption $\frac{d\sigma}{dx} = \frac{dn_1}{dx} = 0$ (For our chosen distance)

$$d\Gamma = -\sigma n_1 \Gamma\, dx$$

$$\int \frac{d\Gamma}{\Gamma} = -\int \sigma n_1\, dx \rightarrow \ln|\Gamma| = -\sigma n_1 x + C$$

This gives us: $\Gamma = \Gamma_0 e^{-\frac{x}{\lambda_m}}$ → Notice how the flux decays exponentially.

where, $\lambda_m = \frac{1}{n_1 \sigma}$ → mean free path.

Collision Frequency

We can get upto shenanigans now that we have the mean free path. We are defining two terms: τ_m → collision time, ν_m → collision frequency.

$$\tau_m = \frac{\lambda_m}{v} = \frac{1}{n_1 \sigma v}$$ → velocity of incident particle

Q: What if v is not constant ∵

$$\nu_m = \frac{1}{\tau_m} = n_1 \sigma v$$

In nuclear reactions, there is just one collision. However is we are dealing with the final charged particles, they can collide w. each other multiple times without changing their identity. In this case ν_m represents the average no. of collisions per second made by the incident particle.

This concludes the discussion on Basic Nuclear collisions. We have classically modelled collisions of microscopic particles using the terms cross-section, mean free path and collision frequency.